

From Adversarial Attacks to Robust Classifiers: A Study in Social Media Spam Detection

Abstract—The rise of social media platforms has transformed communication but also introduced challenges such as detecting spam and malicious content. Adversarial attacks, which subtly manipulate input data, pose a significant threat to the effectiveness of machine learning-based spam detection systems. This study addresses these challenges by constructing a novel dataset of adversarially attacked social media messages using synonym replacement, character-level perturbations, and spammy word insertion techniques. Seven machine learning models, including Logistic Regression, Random Forest, and Gradient Boosting, were evaluated for their performance in detecting spam under adversarial conditions. Logistic Regression achieved the highest accuracy of 99%, demonstrating robustness and reliability in spam detection. This work underscores the critical importance of designing resilient spam detection systems and provides a benchmark dataset and methodology for future research in adversarial machine learning.

Index Terms—Adversarial attacks, Social media spam detection, Machine learning models, Robust classification

I. INTRODUCTION

The exponential growth of social media platforms has made online interactions an integral part of daily communication. However, this expansion has also led to a proliferation of spam, phishing attempts, and malicious content that compromise user security and degrade platform integrity. Social media spam is designed to deceive users into engaging with fraudulent links or disclosing sensitive information, presenting challenges to individual users and platform administrators [1].

Traditional spam detection systems rely on machine learning models trained on historical spam patterns. While effective against static and well-understood threats, these systems struggle to adapt to the evolving tactics of adversarial attacks. Adversarial attacks involve subtle manipulations, such as altering specific words or introducing typographical errors, designed to deceive machine learning models without being detected by human users. In the context of social media, the informal and dynamic nature of text further exacerbates the challenges of detecting adversarially crafted spam messages [2].

The impact of adversarial attacks extends beyond individual users, undermining the integrity of social media platforms, eroding user trust, and increasing operational costs [3]. Platform administrators are forced to deploy resource-intensive methods to detect and mitigate adversarially modified content, straining computational resources and slowing system performance [4]. As adversarial techniques evolve, attackers maintain an edge over detection systems, making long-term defense strategies challenging to implement.

While adversarial machine learning has been extensively studied in domains such as image and audio processing, text-based adversarial attacks, particularly in applications like social media spam detection, remain less explored [5], [6]. The unique challenges of crafting adversarial examples for text due to its discrete nature and the need to preserve semantic coherence have slowed the development of comprehensive studies and publicly available datasets in this domain [7]. This gap underscores the need for further research and resources to enhance the robustness of spam detection systems against adversarially crafted text inputs.

To address these limitations, we constructed a novel dataset of social media messages that includes both original and adversarially modified examples. Adversarial attacks were simulated using synonym replacement, character-level perturbations, and spammy word insertion techniques, mimicking real-world scenarios. This study evaluates the robustness of traditional machine learning models, including Logistic Regression, Random Forest, and Gradient Boosting, against adversarially attacked data. By providing a robust benchmark and highlighting model vulnerabilities, this research contributes to the development of more resilient spam detection systems for social media platforms.

The structure of this paper is as follows: Section II discusses the background and related work, highlighting the challenges posed by adversarial attacks and current solutions in text-based spam detection. Section III elaborates on the methodology, detailing the creation of the adversarial dataset, preprocessing steps, and feature engineering. Section IV presents the experimental setup, evaluation metrics, and results, showcasing the performance of seven machine learning models. Finally, Section V concludes the paper with key findings, implications, and potential directions for future research.

II. BACKGROUND

The rapid expansion of social media platforms has amplified their role as communication hubs but has also escalated the prevalence of spam and malicious activities. Traditional spam detection systems, while effective against static and well-understood patterns, face significant challenges in adapting to the evolving threat landscape characterized by adversarial attacks.

Adversarial attacks are deliberate manipulations designed to compromise machine learning models. Studies have categorized these attacks into three primary types: feature ma-

nipulation, evasion attacks, and poisoning attacks [8]. Feature manipulation alters data features to mislead classifiers, while evasion attacks involve crafting input data that bypass detection systems. Poisoning attacks, on the other hand, aim to degrade model performance by corrupting the training dataset [8].

In the context of social media, adversarial examples are particularly problematic due to the informal language and rapid evolution of trends. Research has shown that text-based attacks, such as synonym replacement, character-level perturbations, and insertion of benign phrases, can significantly compromise spam detection systems [9]. These manipulations are often imperceptible to human readers but exploit the rigid decision boundaries of machine learning classifiers, leading to misclassification [7].

Various defense mechanisms have been proposed to mitigate adversarial attacks. Adversarial training, which augments the dataset with adversarial examples, has emerged as a prominent technique to improve model robustness. Additionally, detection methods focused on identifying adversarial examples before they impact the model have shown promise, although their effectiveness in dynamic social media environments remains limited [10]. Hybrid approaches, combining adversarial training with model enhancements like regularization, have been explored to strengthen spam detection systems against evolving threats [7].

Beyond spam detection, adversarial machine learning has found applications in sentiment analysis, rumor detection, and misinformation filtering. For example, adversarial attacks in sentiment analysis can lead to incorrect sentiment classification, affecting user experience and system credibility [11]. Similarly, adversarial bots on platforms like Twitter complicate the detection of coordinated spam campaigns by mimicking legitimate user behavior [12]. These examples highlight the broader implications of adversarial attacks on social media platforms.

Despite advancements in adversarial machine learning, the field lacks standardized benchmarks and robust datasets for text-based attacks. Existing datasets often fail to capture the nuances of adversarial manipulations in real-world scenarios. This study addresses this gap by constructing a novel dataset of social media messages, incorporating both original and adversarially modified versions. Unlike prior work, this dataset explicitly simulates real-world adversarial scenarios through techniques such as synonym replacement and character-level perturbations. Furthermore, the evaluation of this dataset using traditional machine learning models provides a baseline for understanding the vulnerabilities of current systems.

III. METHODOLOGY

This section outlines the comprehensive approach adopted in this study, starting from the creation of a novel adversarial dataset to preprocessing and feature engineering, and finally evaluating the robustness of multiple machine learning models against adversarial attacks.

A. Dataset Creation

The dataset used in this study is sourced from the Twitter Spam Detection project and consists of 5,169 unique entries categorized into spam and ham messages [13]. Of these, 13% are labeled as spam, while 87% are labeled as ham (legitimate). Although this dataset is well-structured and labeled, its static nature makes it less suitable for evaluating adversarial attacks.

To address this limitation, a novel dataset was constructed by introducing adversarial perturbations to legitimate (ham) messages. These perturbations mimic real-world adversarial scenarios and include techniques such as synonym replacements, character-level modifications, and the insertion of spam-like phrases. The newly enhanced dataset combines the original entries with adversarially modified samples, providing a robust resource for evaluating the performance of spam detection models under adversarial conditions.

B. Data Preprocessing and Feature Engineering

To ensure consistency and uniformity, data preprocessing steps were undertaken. These included handling missing values, removing duplicate entries, and standardizing text formats. Following this, Exploratory Data Analysis (EDA) was performed to investigate the underlying structure of the dataset, focusing on distributions, patterns, and correlations.

Feature engineering was conducted to extract meaningful text-based attributes that could improve model performance. Features such as text length, word count, average word length, and sentiment polarity were calculated to offer additional insights into the dataset.

Sentiment polarity scores were computed using the TextBlob library. Each message was assigned a polarity score ranging from -1 (most negative sentiment) to 1 (most positive sentiment), with 0 representing neutral sentiment [14]. For instance, messages containing strongly positive words such as "amazing" or "excellent" received polarity scores closer to 1, while those with negative terms like "bad" or "poor" were closer to -1. This analysis helped identify the overall sentiment trends within the dataset, which leaned towards neutral-positive sentiment, as shown in Figure 1.

Figures 1 and 2 illustrate the polarity distribution and text length distribution, respectively.

The polarity distribution (Figure 1) demonstrates a strong skew towards neutral-positive sentiment, indicating that the majority of messages contained either neutral or slightly positive wording. A smaller proportion of messages showed extreme positive or negative sentiment, as evidenced by the sparse occurrences near the polarity extremes of -1 and 1.

The text length distribution (Figure 2) reveals that most messages have relatively short lengths, with the word count heavily concentrated below 50 words. However, a few outliers extend towards higher word counts, reflecting variability in the dataset's message lengths. These insights highlight the textual characteristics of the dataset, providing valuable information for model development and tuning.

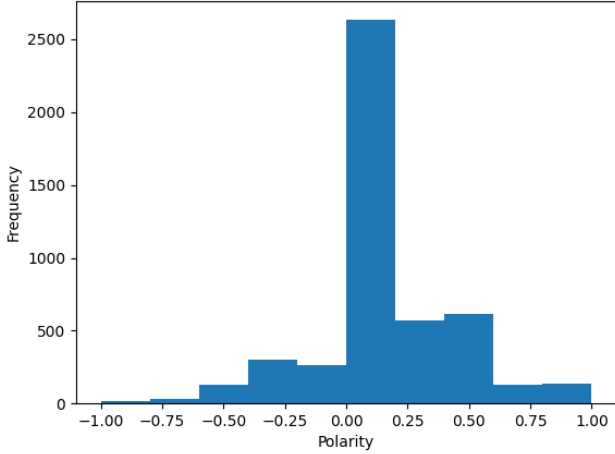


Fig. 1. Polarity distribution of the dataset, demonstrating skewness towards neutral-positive sentiment.

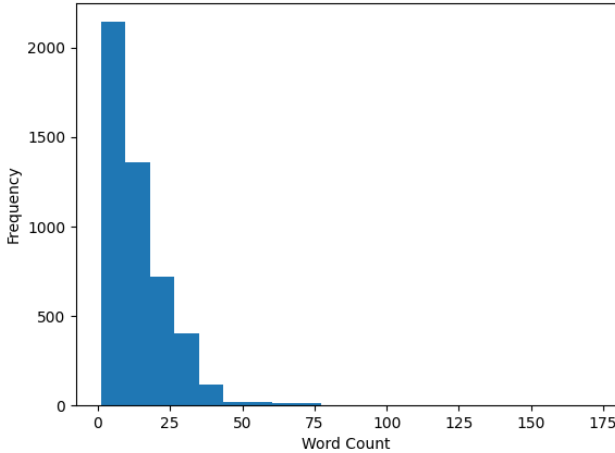


Fig. 2. Text length distribution of the dataset, showing the variability in message lengths.

C. Adversarial Attacks on the Dataset

Adversarial attacks were introduced into the dataset to test the robustness of the model against manipulated text.

As outlined in Table I, three techniques were applied: synonym replacement, character swapping, and spammy word insertion. Synonym replacement substituted specific words with contextually similar synonyms to mimic subtle spam manipulations. Character swapping introduced typographical errors by randomly switching adjacent characters, while spammy word insertion added common spam-like phrases to benign messages. Each message was processed in two forms—original (labeled `is_attacked = 0`) and adversarially modified (labeled `is_attacked = 1`). The resulting dataset, with columns for the

TABLE I
DATASET CREATION STEPS WITH ADVERSARIAL ATTACKS

Step	Description
Dataset Preparation	Begin with benign (non-spam) social media messages free of spam-like characteristics.
Adversarial Attacks	Apply three techniques: Synonym Replacement (replace words with synonyms), Character Swapping (randomly swap adjacent characters), and Spammy Word Insertion (insert spam-related phrases).
Dataset Labeling	Store each message twice: original (<code>is_attacked = 0</code>) and attacked (<code>is_attacked = 1</code>).
Final Dataset	Two columns: message (original or attacked) and <code>is_attacked</code> (binary label).

message text and attack status, provided a challenging test for the model by simulating real-world spam evasion strategies. These adversarial examples helped evaluate the model’s ability to differentiate between genuine and manipulated content effectively.

D. Dataset Balance and Word Cloud Analysis

The newly generated dataset achieves perfect balance in the `is_attacked` labels, with an equal number of samples for each class: 0 (non-attacked) and 1 (attacked), each has 11,144 instances. This balance ensures that the model training is unbiased and not skewed towards either class, enhancing its ability to generalize well across both original and adversarially attacked messages.

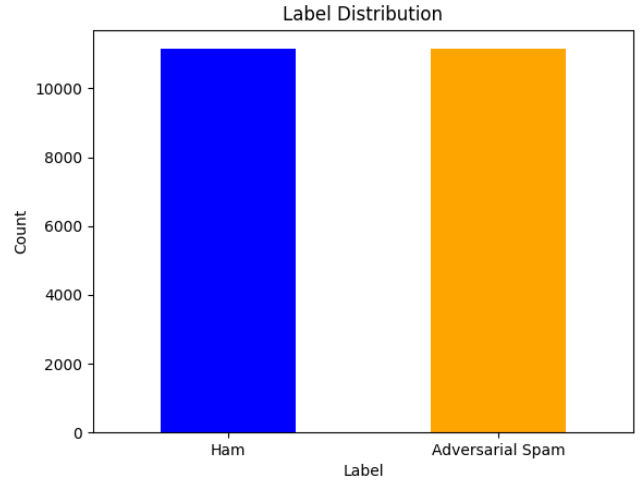


Fig. 3. Label distribution of the dataset, showing a perfect balance between attacked and non-attacked samples (11,144 each).

To further analyze the characteristics of spam-like content, a word cloud was generated from the adversarially attacked messages. The prominent words in the word cloud reveal patterns typical of spammy messages, with terms like “offer,” “free,” and “discount” standing out. These words highlight the spammy nature of the inserted phrases used during adversarial attacks, effectively mimicking real-world spam strategies. This

visualization underscores the effectiveness of the adversarial techniques in enriching the dataset with realistic spam-like attributes, providing a robust challenge for the spam detection model.

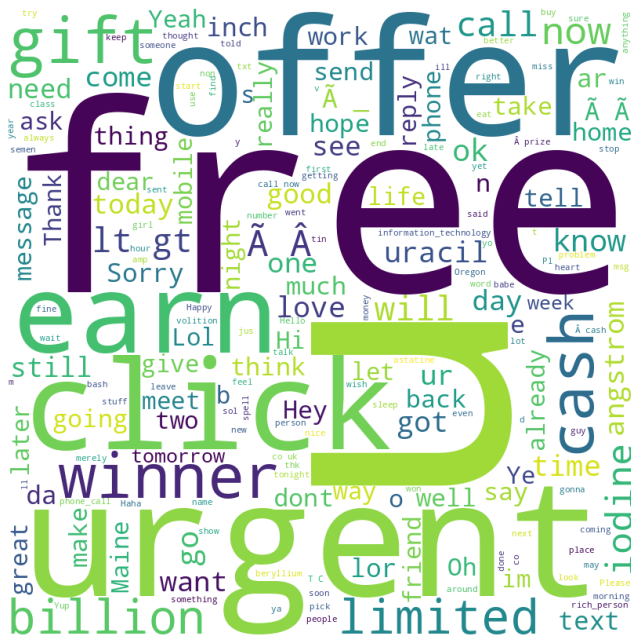


Fig. 4. Word cloud of adversarially attacked messages, highlighting commonly inserted spammy words such as "offer" and "free."

E. Machine Learning Models

After creating the dataset with adversarial attacks, the next step was to train and validate the machine learning models. We evaluated seven traditional machine learning models using the Scikit-learn library in Python. These models were chosen for their diverse methodologies and their applicability to classification tasks involving adversarially attacked data [15]. Below, we describe the selected models and their relevance to this study:

- **K-Nearest Neighbors (KNN):** A simple and intuitive algorithm that predicts the class of a message based on the majority label among its nearest neighbors. KNN is effective for small datasets but can struggle with high-dimensional data.
- **Logistic Regression:** A linear model that utilizes a log-odds function to establish decision boundaries. Its simplicity and interpretability make it a popular choice for binary classification tasks.
- **Support Vector Machine (SVM):** Constructs an optimal hyperplane for class separation, making it well-suited for nuanced classification tasks and capable of handling both linear and non-linear boundaries.
- **Decision Tree:** A tree-based model that splits data recursively based on feature values, providing clear decision

rules. While prone to overfitting, it can effectively handle structured and categorical data.

- **Random Forest:** An ensemble of decision trees that aggregates predictions to improve generalization and reduce overfitting. This method is robust against noise and can capture complex data patterns.
- **Bagging Classifier:** An ensemble learning method that trains multiple base estimators on different subsets of the data, reducing variance and enhancing robustness against overfitting.
- **Gradient Boosting:** A sequential ensemble technique that builds weak learners iteratively, optimizing residual errors at each step. This method is particularly effective in handling imbalanced data and complex relationships.

These models were selected to explore a mix of simple, ensemble, and boosting techniques, providing a comprehensive analysis of their methodologies and applicability. Training and validation were performed on the balanced dataset, ensuring consistent conditions for all models to assess their effectiveness against adversarial spam.

IV. EXPERIMENTS AND RESULTS

The experiments were conducted on Google Colab using a T4 GPU for efficient model training and evaluation. The implementation leveraged the scikit-learn library for building and evaluating machine learning models. The computational environment provided the necessary resources to handle the dataset and execute the adversarial attacks effectively.

A. Evaluation Metrics

To assess model performance, standard evaluation metrics such as precision, recall, F1-score, and accuracy were used:

$$\begin{aligned} \text{Precision} &= \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \\ \text{Recall} &= \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \\ \text{F1-Score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{Accuracy} &= \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}} \end{aligned}$$

These metrics provided a comprehensive understanding of the model’s performance. Precision measured the ratio of correctly predicted positive observations to the total predicted positives, while recall evaluated the ratio of correctly predicted positive observations to all actual positives. The F1-score, being the harmonic mean of precision and recall, balanced the trade-off between the two, and accuracy indicated the overall correctness of the predictions.

B. Hyperparameters

Hyperparameters are critical in defining the performance and behavior of machine learning models. Unlike parameters learned during the training process, hyperparameters are set

before training begins and influence how the model learns from the data. Proper tuning of hyperparameters is essential to achieve a balance between underfitting and overfitting, ensuring the model generalizes well to unseen data.

Table II summarizes the hyperparameters used for each model in this study.

TABLE II
HYPERPARAMETERS FOR MACHINE LEARNING MODELS

Model	Hyperparameters
K-Nearest Neighbors	n_neighbors=5, metric='minkowski', p=2
Logistic Regression	solver='lbfgs', max_iter=100
Support Vector Machine	kernel='rbf', C=1.0, gamma='scale'
Decision Tree	criterion='gini', max_depth=None
Random Forest	n_estimators=100, criterion='gini', max_depth=None
Bagging Classifier	base_estimator=DecisionTreeClassifier, n_estimators=10
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3

C. Results

Table III presents the performance metrics for Ham and Adversarial Spam across the different machine learning models. Among all the models, **Logistic Regression** achieved the highest accuracy of 0.99 and consistently high precision, recall, and F1-score for both Ham and Adversarial Spam. Its simplicity and robustness make it an excellent choice for binary classification tasks in this context.

TABLE III
PERFORMANCE METRICS FOR HAM AND ADVERSARIAL SPAM FOR DIFFERENT MACHINE LEARNING MODELS

Model	Label	Precision	Recall	F1-Score	Accuracy
KNN	Ham	0.55	0.95	0.70	0.58
	Spam	0.82	0.22	0.34	
Logistic Regression	Ham	0.99	0.99	0.99	0.99
	Spam	0.99	0.99	0.99	
SVM	Ham	0.96	0.95	0.96	0.96
	Spam	0.95	0.96	0.95	
Decision Tree	Ham	0.95	0.99	0.97	0.97
	Spam	0.99	0.95	0.97	
Random Forest	Ham	0.97	0.99	0.98	0.98
	Spam	0.99	0.97	0.98	
Bagging Classifier	Ham	0.95	0.99	0.97	0.97
	Spam	0.99	0.95	0.97	
Gradient Boosting	Ham	0.99	0.95	0.97	0.97
	Spam	0.95	0.99	0.97	

V. CONCLUSION

This research demonstrates the critical need for adversarially robust spam detection systems to tackle the evolving challenges in social media communication. By constructing a novel dataset enriched with adversarially modified messages and evaluating the performance of seven diverse machine learning models, this study provides a comprehensive analysis of model vulnerabilities and strengths in adversarial contexts. Logistic Regression emerged as the most effective model,

achieving a remarkable accuracy of 99% while maintaining high precision and recall for both spam and adversarially attacked messages. These findings emphasize the necessity of designing and deploying robust classifiers to ensure the security and reliability of social media platforms.

Future research can build upon this work by exploring deep learning techniques to further improve classification accuracy and robustness against more complex adversarial strategies. Additionally, integrating real-time detection mechanisms and adaptive adversarial defense frameworks can address dynamic and evolving threats in social media environments. The insights and benchmarks provided by this study lay a strong foundation for the development of next-generation spam detection systems, capable of withstanding adversarial manipulations and safeguarding the integrity of online communication platforms.

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